

CSC 281: Discrete Structures, Spring 2026
Course Syllabus
Department of Computer Science
California State University, Dominguez Hills

General Information

Instructor: Dr. Ali Ash. Jalooli

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Office: NSM A-135

Class Time: TuTh 1:00 PM - 2:15 PM

Office Hours: Th 2:30 PM - 4:00 PM by appointment

Class Location: WH F144

Course Website: <https://canvas.csudh.edu>

Prerequisites

CSC 121 and MAT 153

Required Textbook

Essentials of Discrete Mathematics (3rd Edition), by David J. Hunter. ISBN-10: 1284056244 ISBN-13: 978-1284056242

Course Description

This course introduces fundamental mathematical structures and logical principles that form the foundation of computer science. Topics include formal logic, proof methods, set theory, functions, relations, sequences, recursions, inductive proof, counting, selections, arrangements, etc. Such topics will be introduced with an emphasis on applications in computer science. Students will be required to write programs using programming languages to deepen their understanding of lecture materials.

Course Learning Objectives

Upon successful completion of this course, students will be able to:

- Use formal logic and proof methods to analyze and prove mathematical statements and arguments.
- Model mathematical relationships using sets, functions, and relations.
- Solve problems about arithmetic and geometric sequences.
- Write recursive relations and use the technique of induction to prove facts about recursively defined objects.
- Use counting techniques (combination and permutation) to analyze discrete problems.

Office Hours

Office hours are by appointment only and will be held online (via Zoom) or in-person, depending on your preference. The Zoom's waiting room feature will be used for pre-scheduled office hours. If the scheduled office hours don't align with your schedule, let me know, and I will find a suitable time. Note that, we will utilize the same meeting ID (81440581046) and Password (563815) for all of the office hours sessions throughout the semester.

Course Policies

Modules: We are making extensive use of modules to efficiently structure the course content on a weekly basis. These modules serve as containers for various learning materials, including lecture notes, essential dates, and other relevant resources. It is crucial that you thoroughly review each module promptly upon its posting and diligently complete all the tasks and items within a module before proceeding to the subsequent one. It is essential to adhere to the prescribed order of modules as they are presented to ensure a seamless learning experience.

Expectation: My expectation for you is to be engaged and an active participant in the learning process by participating in class discussions. It is important to get started right away and to maintain the pace outlined in your course materials. Also, see below.

Attendance: The attendance policy is rigorously enforced, and inadequate participation and a lack of engagement can have detrimental consequences on your final grade. Absences are required to be discussed with the instructor. Please note that it is your responsibility to drop or withdraw from the course if you are no longer participating. The instructor is not obligated to drop students who are no longer actively involved in the class.

Assignments: Assignments are a central part of this course and play a major role in your learning process. It is important and expected that all assignments are completed and submitted on time. Assignments will include both:

- **Mathematical/problem-solving tasks:** such as logic translations, truth tables, proofs, set operations, and graph theory exercises.
- **Programming tasks:** primarily in **Java**, where you will implement discrete structures concepts computationally. Your source code should always be clearly written and appropriately commented, and your written solutions must be well-organized and justified.

Assignments will be distributed through **Canvas** and must be submitted electronically by **11:59 PM on the specified due date** (unless otherwise stated).

Late Submissions: Late submissions are generally not accepted, as timely completion is essential for keeping pace with the course material.

Exams: There will be two in-class exams, a midterm and a final. The specific dates and instructions for the exams will be announced at a later time. It is important to highlight that in order to successfully pass this course, students are required to take both exams. No makeup or early exams will be administered.

Quizzes: There will be 2 quizzes, 1 leading up to each exam. Quizzes are practice for the exams.

Evaluation

Grading: The course grade is determined by the following components:

Assignments	20%
Quizzes	20%
Exam 1	30%
Exam 2	30%

Grade Scale: Final grades will be assigned according to the following scale:

Grade	Score Range	Grade	Score Range
A	≥ 94	C	[65, 70)
A–	[90, 94)	C–	[60, 65)
B+	[85, 90)	D+	[55, 60)
B	[80, 85)	D	[50, 55)
B–	[75, 80)	F	[0, 50)
C+	[70, 75)	NA	–

Communications: [Course website](#) is the main source of information about the course. Most course materials such as syllabus, handouts, discussions, assignments, solutions, and announcements will be posted there. Therefore, make sure to regularly check the course website.

University email should be used for communications. Email from other accounts could end up in spam and not being responded. Note that, when sending an email, include the course number in the subject of the email (i.e., CSC 281).

Academic Integrity

Academic integrity is of central importance in this and every other course at CSUDH. You are obliged to consult the appropriate sections of the University Catalog and obey all rules and regulations imposed by the University relevant to its lawful missions, processes, and functions.

- All work turned in by a student for a grade must be the students' own work. Plagiarism and cheating (e.g. stealing or copying the work of others and turning it in as your own) will not be tolerated and will be dealt with according to University policy. The consequences for being caught plagiarizing or cheating range from a minimum of a zero grade for the work you plagiarized or cheated on, to being dropped from the course.
 - Do not show another student a copy of your homework or projects before the submission deadline.
 - Do not email your project to another student, even if they promise they will not copy it.
 - The penalties for permitting your work to be copied are the same as the penalties for copying someone else's work.
 - If you choose to do your work on your computer, make sure that your computer account is properly protected. Use a good password, and do not give your friends access to your account or your computer system.

- Do not leave printouts, or thumb drives around a laboratory where others might access them.
- **Copying = “copying and then modifying”:** You can copy material from other sources only if the instructor gives you clear permission to do so. When this syllabus or the instructor says “copying” you should also interpret this to mean “copying and then modifying.” If you copy material without permission and then modify it (to make it more difficult to detect, or for other reasons), you could still be reported.
- **Cite all materials that you use:** If you use a resource to help you complete an assignment, you must cite that material. If you used a website to help you solve a compiler error, you should leave a comment (example: “I used information on web page XYZ to help me solve a compiler error”). If the instructor gives you permission to reuse code, you must indicate which section of code you copied and where you copied it from (example: “I copied this function from web page XYZ and made some small modifications to it”).
- **Have questions?** If you have questions or concerns it is much better to talk to the instructor while you are working on the assignment than to hide what happened and hope the instructor doesn’t notice.

IMPORTANT: I will actively look for cheating with software that detects plagiarism and collaboration in English text and computer code. Ideally, your solutions will not match any of the other solutions that it is compared to. Consequences: Any violations will result in an F grade in the course and university disciplinary action.

Zoom Policies

As mentioned before, we use Zoom for office hours. Therefore, make sure to download and install the Zoom client for such purposes. You can download Zoom [Here](#). The following are office hours policies for our meetings with Zoom. Please read carefully, these policies are effective immediately and apply for the remainder of the semester. All students are expected to adhere to the policies.

Knowing Your Responsibilities

CSUDH provides you with a wide variety of academic assistance and personal support, but it is up to you to know when you need help and to seek it out. It is your responsibility to keep informed and to obey the rules, regulations, and policies which control your academic standing and your life as a CSUDH student. Meeting deadlines, completing prerequisites, and satisfying the degree and certificates requirements, as found in the curriculum guides in this catalog, are all part of your duties as a student. Consult this catalog, the college, and school announcements and the schedule of classes for the information you need. Watch for official announcements.

Accessibility at CSUDH

Access to publications, instructional material, computer software, hardware, and electronic information, as well as access to the campus, are critical for the educational and career achievement of all persons. CSU Dominguez Hills seeks to enable that access with this directory of information and services. The policy of the CSU is to make its programs, services, and activities accessible to

students, faculty, staff, and the general public who visit or attend a campus-sponsored event, with disabilities.

Disabled Student Services, also known as the [Student Disability Resource Center](#) (SdRC), requests students to bring any issues and concerns related to service delivery or access to the University's buildings, programs, and activities to our attention. SdRC will either address the issue and concern itself or forward it to the appropriate University department. Your willingness to make us aware will enable the SdRC program and campus to better meet your needs.

Special Needs

Online courses are required to meet ADA accessibility guidelines. This means that all aspects of the online learning experience are accessible. Please let me know if you have adaptive software and hardware to assist you with taking this course or if you have any specific needs I should be aware of. The [CSUDH Student Disability Resource Center \(SdRC\)](#) is available to assist you during this course. The SdRC is available at (310) 243-3660 and can be reached by email at dss@csudh.edu.

Technical Help

If you need technical help, refer to the following resources:

Login Issues: For log-in issues related to Canvas, Toromail, and MyCSUDH, contact the [IT Help Desk](#) at (310) 243-2500, option 1. You can also create an [online service ticket](#) for login support. The IT Help Desk also offers walk-in support. Visit the first floor of the library (north), C-108, for in-person help.

Password Resets: CSUDH offers an easy, self-service password reset service at <https://password.csudh.edu/>. For additional assistance with password resets, contact the [IT Help Desk](#).

Canvas Issues: For issues or questions with Canvas, contact the CSUDH Canvas Support line at (310) 243-2500. You can also create an [online service ticket](#) for Canvas support.

Need Help with Using Canvas? If you are new to Canvas or unfamiliar with a specific feature of Canvas, [CSUDH Academic Technology](#) offers a series of PDF and video-based tutorials. Visit the [CSUDH Academic Technology Tutorials](#) for more information.

Student Academic Appeal Process

Authority and responsibility for assigning grades to students rests with the faculty. A grade appeal is permitted when a student can show clear evidence that a grade was contrary to procedures as specified in the course syllabus, was based on prejudice, was capricious, or was the result of computational or clerical error. The presumption is that the grades assigned are correct until there is a clear demonstration otherwise. The burden of proof is heavy, and it rests with the student who is appealing.

Incomplete

The grade of I is intended for the rare circumstance when a student who has been successful in a

class has an unexpected event occur shortly before the end of the class. I will not consider giving a student a grade of I unless the following three conditions have been met.

- It is within two weeks of the end of the semester.
- The student has a grade of C or better in the class.
- The reason that the student cannot complete the class is properly documented and compelling.

Americans with Disabilities Act

adheres to all applicable federal, state, and local laws, regulations, and guidelines with respect to providing reasonable accommodations for students with temporary and permanent disabilities. If you have a disability that may adversely affect your work in this class, I encourage you to register with Disabled Student Services (DSS) and to talk with me about how I can best help you. All disclosures of disabilities will be kept strictly confidential. NOTE: no accommodation can be made until you register with the DSS. For information call (310) 243-3660 or to use the Telecommunications Device for the Deaf, call (310) 243-2028 or go to: <http://www4.csudh.edu/dss/>.

Behavioral Standards

Disruptions of the class will not be permitted. Behavior that persistently or grossly interferes with classroom activities is considered disruptive behavior and may be subject to disciplinary action. This applies to online behavior as well. Such behavior inhibits other students' ability to learn and an instructor's ability to teach. The instructor may require a student responsible for disruptive behavior to leave online class pending discussion and resolution of the problem and may also report a disruptive student to the Student Affairs Office (WH A-410, 310- 243-3784) for disciplinary action.

Examples of disruptive behavior include:

- Allowing a cell phone or pager to repeatedly beep audibly while microphone is unmuted.
- Playing music or computer games during class in such a way that they are visible or audible to other class members.
- Exhibiting erratic or irrational behavior.
- Behavior that distracts the class from the subject matter or discussion.
- Making physical or verbal threats to a faculty member, teaching assistant, or class member.
- Refusal to comply with faculty direction.
- Performing a distracting repetitive act such as talking/chatting to others
- Email harassment

The contents of course material provided by the instructor, the links contained therein directly and indirectly, and the contents of the said links, are copyrighted. They are provided exclusively for non-profit educational use by the students currently enrolled in this course and for the duration of this semester. No other use or any use by others is allowed without authorization of the instructor

in this course and copyright holder(s). No videotaping or recording without instructors' prior permission is allowed in class.

For more details on academic integrity, please check the [Academic Integrity Policy of CSUDH](#)

Basic Needs Resources for Students

If you are struggling financially, can't afford healthy, nutritious food, or need stable housing, take advantage of these resources on campus. <https://www.csudh.edu/student-services/food-shelter-resources/food-resources>.

- DH Eats App – Look for it in the Apple App Store
- CalFresh – Financial aid for food to help stretch your grocery budget. Email calfresh@csudh.edu with your eligibility questions. Or make an appointment: <https://dh-calfresh.appointlet.com>
- Housing Referrals – Contact the Basic Needs Coordinator to go over temporary housing solutions. mkirk@csudh.edu
- Basic Needs Food Distribution Events – The basic needs program is providing (4) meals and fresh produce to students every other Thursday. Event registration link will be sent out via toromail and iToros app.
- Farmers' Market – Turn 10 CalFresh dollars into \$20 worth of fresh produce through the DH Farmers' Market's "Market Match" program: Tues 10-2 in the Sculpture Garden.
- Toro Food Pantry – Another nutrition assistance program available to any CSUDH student. They also stock toiletries and menstrual products.

Tentative Schedule

The following table provides a summary of what we do each week. This is a tentative plan and is subject to change. For an updated plan, please also refer to the Orientation Module on your course website.

Week	Topic	Week	Topic
1-2	Formal Logic, Propositional Logic	8-9	Midterm Exam + Sequences, Recurrence Relations
3-4	Predicate Logic, Methods of Proof	10-11	Closed-Form Solutions, Proof by Induction, Basic Counting
5-6	Sets, Functions	12-13	Combinations, Permutations, Time Complexity and Big-O
7-8	Relations and Equivalence	14-15	Three Algorithm Types, Final Exam Review and Exam

Important Dates

Please refer to the university academic calendar at <https://www.csudh.edu/academic-affairs/academic-calendar/> for a complete and updated list.